

Farm IoT API - State 1 - Needs Assessment

This is a working document for the needs assessment stage of the common API project.

The original grant proposal is [here](#).

See also the [Notes doc](#) which has random notes that need saving but don't fit into a "Needs Assessment".

Prior to working on this we need a quick understanding of who is going to do what for the project as a whole..

Please feel free to edit this document, or add comments and suggestions for discussions.

Assumptions

This is a set of working assumptions - they may not (yet) be correct.

There are two common cases

- Ongoing - sensors that measure and send data
- Static - e.g. Lab - devices that send single readings on request. For example a soil test.

In most cases the devices will talk to some kind of backend using their own protocols.

Because the devices are small and optimized for example for bandwidth, it will - in most cases - be easier to have the platform's backend talk the API. E.g. Frugal-IoT 'logger' pushing data to LiteFarm, rather than a Frugal-IoT or OurSci etc device talk direct to LiteFarm.

Divya: It also helps to map the user journey. Does a user with a sensor register or FrugalIoT first? will the login be separate or do you want to have an sdk within LiteFarm/ FarmOS for registration (so there's no separate tool visible to a user, basically FrugalIoT itself is the backend for sensor related stuff for farm management). Then will FrugalIoT have a UI? What a user can or cannot do in FrugalIoT? I'm happy to help figure these out

Mitra: These are good points.

Frugal IoT will always have a UX, but ideally I would see that for LiteFarm users, the entire UX comes from LF.

We probably need to expose the FrugalIoT UX to LiteFarm users as an advanced feature - especially as a way to access new FrugalIoT function in the period between us implementing it, and LiteFarm integrating it. This could be as simple as embedding FIOT's UI as an embedded window with LF authenticating with FIOT.

Areas of needs assessment

This API needs to cover:

- User stories - to set the rest into context.
- Data that needs to be transferred - what kind of data are we moving
- Representation of that data - how do we express that data, including any additional metadata
- How we transmit one or more items of data - e.g. opening a session, authenticating etc etc
- Setup / Admin / Changes, for example when a new device gets added, or a new farm
- Security and Privacy constraints
- Balancing efficiency vs Ease of implementation - “configuration document” was mentioned, but unclear in what context.

Below is one section for each of these.

Greg: I think mapping existing APIs for the list of major product types is really helpful. Just seeing how data is sent today, side by side, would help me. Not sure if that's helpful for others but just a thought.

Mitra: Do you have access to such info? I don't. And there is not enough budget in this process to do a review of devices that might or might not ever get connected.

Greg: I could give it a try if you can identify a list of most common farm connected tools (greenhouse controllers, field monitors, etc.)

Mitra: I don't have such a list - in most parts because for the people I'm engaged with in Resource Constrained environments none of these tools are accessible.

Typical example - was chatting with someone working to improve quality of coffee production in Indo, had been quoted around US10k for a drying shed control system. While he could afford that as a one-off experiment, its not a replicable solution. Similarly greenhouse systems have several zero's too many on the price tag. I'm happy to actively include any that people are using, but not going to focus on seeking out ones that are not going to work in Resource Constrained environments

User Story

A collection of User Stories. In the following - to keep it simple I've used the following specifics as place-holders

- LiteFarm - for any platform
- FrugalloT - for any device platform with continuous data
- OurSci - for any device platform with point-source data
- Farmer - a typical user of a platform, has control of land, with devices on it
- Researcher - someone accessing data but without control of the specific devices (or land)

Greg: as real as possible. In a perfect world, these are things people are actually doing today (with funding or self-funded). Are there any such efforts happening?

Mitra: Unsure of your question - this section - the sections below - (Registration to Diagnosis) are example stories. This list here is just to make the stories more readable. I moved FieldKit to a section at the bottom.

Greg: Unsure of your question - this section - the sections below - (Registration to Diagnosis) are example stories. This list here is just to make the stories more readable. I moved FieldKit to a section at the bottom.

Mitra: Yes - greenhouse and post-harvest value add are the areas we've identified as well.

Registration of new device

- An existing farmer on (e.g.) LiteFarm
- Accesses an "Add Device" menu
- Enters the device ID as provided by the device supplier
- Enters a GPS location, or locates it on a map - this is manual, most devices won't have GPS chips.
 - LiteFarm communicates with (e.g.) Frugal-IoT and registers the device
 - Some form of authentication may happen to prove the user is legitimate to access the device.
 - Frugal-IoT records that data for this device should be sent to LiteFarm
 - Frugal-IoT probably has to tell LiteFarm what kind of device it is, and any parameters.
- Data flows from Frugal-IoT to LiteFarm
 - Nothing in this user-story says whether information flows immediately, or periodically, and whether its pull or push.
- LiteFarm displays the data to the user
 - Its up to LiteFarm how to display this - as a gauge or text or a graph

Divya: When a user adds their device serial ID or some unique identifier, we can make the user-device association then. If another user comes and tries to add the same serial ID, we can prevent that. But there is no direct way to say that this device definitely and only belongs to this particular user.

Mitra: That probably works - the odd case would be if the user registered in LiteFarm, but the device was already registered in FrugalIoT (potentially by someone else)..

Divya: We probably should think of webhook and an API for automatic vs on-demand data. To start with, we could do just one. Is it possible to define a webhook only if a threshold is reached
Eg: push only if soil moisture is below a certain level (actionable and urgent) but pull for all other use cases (mostly for insights as opposed to immediate actionable)

Mitra: From FrugalIoT's perspective, actioning a webhook (event driven POST) is relatively easy, defining the UI for them is more complex as it would involve a UI on LiteFarm, then a way to communicate that to FrugalIoT and then (the easy part) a way to send the data back. IMHO Its worth thinking it through, and defining it, even if it isn't implemented in V0.1

Historical data

- A farmer on (e.g.) LiteFarm wishes to get historical data for their devices
- LiteFarm queries (e.g.) FrugalIoT for data e.g. by date range
- FrugalIoT returns that data in a form LiteFarm can understand.
 - Note that LiteFarm already knows about the device (e.g. that its a temperature and humidity guage) based on the registration process above.
- LiteFarm displays a historical graph, or some analysis of the data.

Divya: you might want to consider a cut off date like 3-6 months max beyond which you purge, just so that there isn't bloat

Mitra: Yes - we'll have something like that on the Frugal-IoT side, for now, no limit, but if start hitting high volumes then will need a way to cost-recover (which should be low, the amount of data is trivial compared to images and video).

Point Source Data

A user has a Reflectometer and wants to take leaf chlorophyll measurements (proxy for Leaf N) 6 weeks after planting to determine the amount of topdress fertilizer to add to a crop. The user creates a SurveyStack survey asking key questions (field ID, GPS location, crop, leaf being measured, etc) and adds a series of measurements. For example, the protocol may call for measuring 5 plants in the field and measuring 3 leaves of different growth stages on each plant.

In the field, the user follows the instructions in the survey, entering metadata at each of the 5 plants being measured and then scans the leaves for chlorophyll content.

Once the survey is complete, the results are submitted to SurveyStack. From there, Litefarm can access the results via the SurveyStack API.

Litefarm can process the leaf N data and use its decision support tools to provide the producer with topdress fertilizer recommendations.

Greg: this is interesting... I would kind of assume we'd want a standard that can go directly to whoever wants it... ie to Litefarm, or FarmOS, or Home Assistant, or whatever... Surveystack is great but is an intermediary.

Maybe we should be digging more into how our data is structured... and that can help identify at least the key components of what a localized farm standard can look like if that's helpful?

Maybe actually I can be a case example... we could sit down and walk through the data flow process and that may help.

Mitra: Its a valid question. but I don't believe you can build the APIs into the devices, see the section I added below to address this ... "Why this is **not** a device direct to backend API"

Not sure I understand your last paragraph, Dan added the OurSci section - is that what you mean by "the data flow process" or could you expand on it.

Researcher

- A researcher wishes to access historical data on a range of devices
- Some process will be required to authenticate that the researcher is allowed access to this data.
 - Its unclear to me how to do this?
 - One option is that
 - the researcher is authenticated to LiteFarm, so LiteFarm knows they have access to the devices,
 - FrugalloT has (during registration) determined that at least one user of LiteFarm has rights to data from each of the devices requested,
 - FrugalloT trusts LiteFarm to apply appropriate policies (for example have the user's consented as a condition of use of LiteFarm?)
- Either data is packaged up and returned that matches an exact query - in which case we need some version of a query language (not that hard)
- OR LiteFarm has access to "Historical Data" as in the previous User Story, and FrugalloT sends back that data, which LiteFarm processes for the researcher.

Notes incorporated from Comments:

Divya: The key is anonymized data on a 'range of devices'. I cannot think of use cases involving aggregate data on sensors alone. I think it makes sense to club it with some farm management practices data. I think it needs some thinking, we could park research related solutions for later?

Note there is an assumption that devices don't hold data long-term, they transfer as soon as practical and then that data is held on something that can do permissioning (e.g. FrugalloT or LiteFarm).

Irrigation Control - simple

- The farmer looks at the data on their display and decides to turn on the irrigation for 50 minutes.
- LiteFarm communicates a control message to FrugalloT
- FrugalloT turns on irrigation
- Either
 - LiteFarm communicates again after 50 mins << more likely as could be sophisticated control
 - FrugalloT turns it off after 50 mins

While FrugalloT can handle a certain set of conditions, I think it makes sense that LiteFarm manages such complex controls.

Divya: Device actions is the next step after data transmission is figured out. We could identify simple use cases for all device actions (turn on/off sprinkler/light/something else)? Can a universal turn on/off api exist for all devices for example? Won't it depend on type of device?

Mitra: I think there are basically three places that control can happen.

a: on the device - Frugal-IoT supports this

b: in the UX (browser) - we support this manually, but not automatically as the user may not be connected at some future time

c: in the server - this is not currently supported in Frugal-IoT

d: In LiteFarm which - as you say - would need to know the device description.

We currently have a complex device-description file It can communicate the difference between say a Relay with just on/off or a device-side control with set-point, hysteresis etc. There is some tech-legacy-debt in the device description file, so I want to clean it up before sharing it !

Irrigation Control - complex (for future thought)

- The farmer decides they want irrigation pump on when sensor 3 shows <50% moisture content and temperature >30°C
- Either LiteFarm sets this up as a trigger event
- Or LiteFarm sends this to FrugalloT

While FrugalloT can handle a certain set of conditions, I think it makes sense that LiteFarm manages such complex controls.

Notes from Comments:

Greg Austic: I would suggest this... I would expect each device publish a spec (when you register) which defines it's parameters. There, there may be a parameter for 'timeLeftOn' or something which you can set to '50' in the initial control message. Is that what you're thinking also or am I in left field?

Mitra: That's how we do it on FrugalIoT, I'm guessing others do similarly.

Greg: the key is that when the device is registered, it publishes a schema containing everything anyone would need to know about how to interact with it IMO. Like the first thing I get is 'whoAmI' or something which contains the rules for interacting with me, what I can do or can't, etc (in code, like json schema or similar)

This has to exist right? I would see if there's groups at Linux foundation or Ag Gateway that have this kind of back and forth figured out...

Mitra: Yes - some kind of device schema, which may be from the device, or from a backend (since many devices don't have enough space to hold a schema).

Note - AFAIK there is no industry standard or widely used practice, for such a schema, we have one at Frugal IoT and we'll propose that - though I'm not attached to using ours versus some better way.

Diagnosis of sensors

- Farmer looks at map
- Map displays location of sensors, and flags dead or alive,
- Farmer interrogates map
- LiteFarm interrogates Frugal-IoT for diagnostic info
- LiteFarm displays to user

Data to be transferred

From Frugal IoT's perspective we commonly deal with Temperature, Humidity, Lux, Soil Temperature, Soil Moisture.

We've seen other similar devices doing: Soil PH, PK,

Weather parameters are also likely - wind speed, pressure, rain etc

This of course should be extensible.

What does OurSci measure?

What data does LiteFarm need?

How do we specify the meta-data around data (is Common Farm part of that ?

This needs to also cover information in the opposite direction (for control/configuration)

Greg: we generate raw data (spectral values), but usually it requires chemometric models to convert that to useful data (like soil carbon, or leaf greenness). Mostly we've done those conversions off the device, which makes sense, but it is less elegant because it requires more stuff to work.

I would love to do more on-device, or something (in your spec) ensure that any additional code that needs to be run is clearly identified and referenced when you register the device. Like "i send this type of data and it looks like this" but also "and you have to go get this code to interpret my

data to get XYZ values". The vast majority of useful sensors require calibration, adjustments, have various adjustment factors or full on models to get meaningful data off them. IMO simple stuff like temp and humidity are the exception not the rule.

Mitra: Agreed - generalizing this is highly unlikely - i.e. LiteFarm can't be expected to implement your model for processing spectrographic data), which is another reason this goes in the device's platform, with pre-processed, standardized values communicated over the API

Representation of that Data

Dan suggested [Common Farm Conventions](#) and someone said that there are ways to add extensions to it?

Extensibility

Whatever representation we use needs to be extensible in two situations

A: A way for collaborating partners to use a new data element not defined in a standard/API. For example, in email by convention any header starting with "X-" is non-standard.

B: A process to extend the standard that is not too cumbersome.

Efficiency

If its device to back-end then we need to be concerned that devices may be on very low-bandwidth connections (e.g. LoRa), but if we presume that the communication is back-end to back-end then efficiency may not (within reason) be a priority.

Readability

Ideally the data should be somewhat human readable to make it easy to debug and implement (e.g. JSON, JSON-LD or YAML)

Greg: I think the CFC is helpful at a high level (ie the entire stream lives in a 'log' object)... but I doubt it's going to help or dictate the exact protocol details. It is JSON Schema, so you could also then use JSON Schema to control / validate down into the machine data as an option for sure, but I can't say that those standards exist. But... if you use JSON in a way that can be schematized, then sure it's as good a system as any. Let's review some real data as examples!

Mitra: Do you have a link to the right place within CFC for some useful data examples ?

Greg: I could be curious to get feedback on this from the forum, providing some specific case examples and data if possible.

Mitra: Sorry, which forum ?

Greg:: sorry - [farmos.discourse.group](#) is probably best or at [forum.goatech.org](#)

Mitra: I'm on [Forum.goatech.org](#) - which is very quiet, not sure where or who to ask there - or even how to phrase the question so its understood in context.

Communication of one or more Data items

We might need multiple communication mechanisms, that communicate the same data, using the same representation.

There are a number of options - (we might not need them all)

- Immediate Push - new data points communicated as soon as available
- Periodic Push - new data points uploaded periodically by the device's platform
- Periodic Pull - the Farm software requests a set of, or a single, value

We could use ...

- MQTT - especially if doing immediate push, but it might not be as easy for Farm software to integrate
- Simple HTTP Post or Get - sending a JSON or YAML file that conforms to the Representation above.
- The format of the packet sent by MQTT or HTTP also belongs here, and we may want to consider the IETF's [RFC 8428](#) standard

Whichever method is used we need to address authentication.

Setup - Installation - Admin - Changes

What are the common admin changes that get made, and how are these communicated. (See the User Story's above, and add more as required)

Example I can think of.

- Add a new "farm"
- Add a "device" to a "farm"
- Configure a device
- Delete a "device"
- Delete a "farm"

Actuation

Are there any actuation applications where the Farm Platform wants to reach out to a device - for example to turn irrigation on/off. Is this a concept in the Farm Platforms?

This concept exists in Frugal-IoT who would be willing to extend it to the API. Divya at LiteFarm has requested this ability

Greg: Tons - greenhouses are the main one. Oreisha is a canadian company that does automation of greenhouses. Also there were two great folks at MOFGA from Canada you should get case examples from also (example data and also just talk with them about this). I'll get their email shortly.

Mitra: Yes , I think we have consensus that this has to include actuation in the API. For example, we just added simple climate automation (temperature sensor driving fan) to a BSF fly cage in Indo, total cost under \$20.

(I Note that contact info was sent in email)

Security and Privacy concerns

A general place-holder for thinking this through. How we deal with authentication, keeping data private etc.

Authentication needs to be able to work with the existing mechanisms of the Device-Platform and/or Farm-Platform. It should not require separate registration processes. See the User Story's above for examples - especially the more complex "Researcher" example

Organizational

What needs do we have for organization going forward - for now I think it can be an informal mailing list, but if the group grows we'll need a process for reviewing changes to the API.

There was another FLOAT proposal for creating a higher level org structure for Open Ag standards, and I'd be happy to collaborate with that as a working group if it makes sense.

Greg: This is moving forward! I'd be happy to talk with you and Juliet about how it has worked so far so you know if and when you'd want to engage with it.

Mitra: Happy to setup a call - late next week or the week after might be better as I'm currently in Indonesia, but back in Australia from Thursday. Can you set this up

Greg: sure - done!

Terminology & Abbreviations

I want to suggest that we agree on a set of terms, so that we can talk across different assumptions. The definitions below may need adjusting. (Please feel free to add the words used to describe this in your own organization).

- Common Farm Conventions (CDC) - a spec for communicating farm data (over JSON-LD?)
- Device: A sensor device in the field or lab, with some kind of communications and sensors. (FrugalIoT:node) (in other places this is called "Node" or "Thing")
- Device-Platform: A platform that devices connect to natively (e.g. frugaliot.naturalinnovation.org) (FrugalIoT: Server)
- Farm-Platform: A platform that provides a UX to farmers.

- Farm: A location with one or more devices, a logical group as far as a “Farm-Platform” is concerned. (FrugalIoT: Project)
- Actuation - the process of the platform sending a message to the Device causing an action e.g. turning on irrigation, or changing a setpoint at which irrigation comes on automatically.
- Control - processes within a platform that make decisions from sensed data and then cause Actuation - Control is outside the terms of this spec as it is internal to a platform, However changing setpoints on control systems operating inside devices may be pertinent to this standard.

Why this is **not** a device direct to backend API

There have been legitimate questions of why this is not an API to run in devices, allowing them to talk directly to any backend. I believe the combination of the following reasons make it impractical to assume this, though the API shouldn't preclude a bigger, more capable, device talking the API directly.

- Most devices don't have IP addresses, they'll typically be either:
 - On temporary NATed WiFi IPs,
 - Intermittently connected via a range of means (including BLE to a phone)
 - Behind some other low bandwidth protocol (like LoRa)
- Devices, are often already fully baked - code blown at device manufacture and in the field.
- Devices typically have low capacity so supporting additional protocols has significant tradeoffs not present with the back-ends. For example most ESP32's have 4Mb of Flash making an app maximum size around 1.5Mb, (you need room for two, in order to do OTA, plus data saving). Many applications are already pushing that 1.5Mb limit (due to code bloat from the RTOS etc).
- Comms protocols are optimized for different low bandwidth situations (including LoRa), there are tradeoffs between bandwidth and generalization, that don't happen between back ends, thus allowing devices to remain optimal connectivity with their own backends, while retaining flexibility backend<>backend.
- Device data is often raw - for example spectrographic info - and needs processing to turn it into something useful. For device constraint reasons (as noted above) this is typically done off-device. The presumption is that the people who know how to interpret it are the device builders (e.g. OurSci for their Reflectometer) and that processed values are what get communicated to the user's platform (LiteFarm).

Specific Needs or discussion from each platform

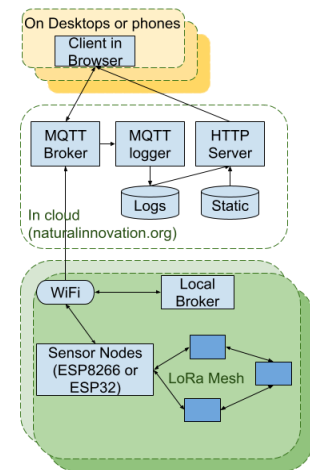
Please add how your platform works with this kind of thing and anything you need that doesn't neatly fit in the topics above.

Frugal IoT

Frugal IoT has the following architecture, it interfaces to external systems from the Logger. So for example with Google Sheets or Firebase it accumulates data for specific nodes, then periodically sends these to Google Sheets or Firebase using that system's protocol.

I imagine it doing something similar for this API - i.e. collecting data and converting to JSON-LD/Common Farm Conventions (CFC) and then uploading via a HTTPS Post or similar. The mapping between Frugal IoT data elements and CFC would be done in the Logger, either as a config file, and/or in code.

Frugal IoT would expose a HTTP GET Interface for requesting historical data, or data sets for researchers etc.



OurSci

The Our Sci Reflectometer takes point soil (SOC) or plant (chlorophyll content) measurements.

1. The scripts (javascript) controlling Reflectometer measurements are added to a SurveyStack survey, which also uses survey questions to generate measurement metadata. There are two types of metadata:
 - a. Device metadata - timestamp, humidity/temp, device ID, firmware version, etc, part of the json output from the device
 - b. Survey metadata - GPS location is a survey question, along with other identifying information (Farmer/Farm name, sample type, etc)
2. In the field, the user completes the survey in SurveyStack and runs Reflectometer measurements.
 - a. Open up the survey in SurveyStack app, answer survey questions
 - b. Connect the Reflectometer to the phone using BT
 - c. Run Measurement - measurement commands sent to Reflectometer (BT)
 - d. Measurement data returned to SurveyStack (JSON).
 - e. Measurement and survey data uploaded to SurveyStack db as JSON when survey "submitted."
3. SurveyStack has a robust API that would allow other platforms to access survey and Reflectometer data.
 - a. Survey metadata will be dependent on the survey object and may not be consistent across different users/surveys.

Mitra: The complexities here confirm what I think is the case for Frugal-IoT as well, that the interface should be Survey-Stack pushing the data through to LiteFarm (&FarmOS)

Mitra: Does this have to be "Pull" (LiteFarm pulling data from SurveyStack) or is SurveyStack able

to push reading to LiteFarm as they are taken.

Greg: surveystack will push. It does this already with FarmOS

Greg: I would love to revisit this whole thing in a more generalized way honestly

@mitra.ardron@gmail.com . I think newer tech and tools could reduce the need for intermediate software (like... where and how we run those scripts to make spectral data into useful outputs) and really generalize it to a wider range of platforms. Maybe out of scope, just noting at least that I think you should try to push for more elegant and simple ideas here and don't follow where our trail leads to spaghetti!

Mitra: If the API makes sense it will support changes in where things are run. This description is the current flow (as posted by Dan), this doesn't get baked into an API, but an API needs to support the edge between SurveyStack and LiteFarm (or between devices and LiteFarm)

If I'm missing something then maybe you could be more specific.

Greg: cool. Yes, this is the current process, but agreed as long as we can effectively document / describe the APIs or connections then it can be generalized.

LiteFarm

Divya: Please add anything you want here.

I will share API request response structure from a previous implementation, might be useful to work off of that.

Use case	Purpose	Request parameters	Response body parameters
When a user registers device on a different platform like FrugalIoT (not on LF)	To get all devices associated with the user	GET Org ID (for LF/FarmOS) userID (specific user)	List of: ID, label, device type, serial num, firmware version, hardware version, isOffline, Isdeployed, owner org name, last requested/seen, {latest position object including timestamp, lat long, altitude, label, environment}
When user registers on LF	We don't have a flow for this	NA	NA
Getting data from sensor	Getting data from sensor	GET We use the organisation primary key, as above, and	Just an example: {

		additionally the unique IDs of the sensor(s) on the platform like FrugalIoT, a start and end time, and a truncation period (e.g. hour, day) which determines how granular the datapoints are.	<pre> "Label" "Label_fallback" "Device_uniqueID" "device_type": " "configurable": false, "firmware_version": "5a23fa5", "hardware_version": "F", "current_zone_id": 220, "data_count": 25, "data": [{ "parameter_category": "Soil Water Content", "unit": "%", "label": null, "parameter_number": 2, "timestamps": ["2025-08-01 00:00:00+00:00", "2025-08-02 00:00:00+00:00",], "Values": 35.15007541236119, 33.55428542722651}], { "parameter_category": "Temperature", "unit": "C", "label": null, "parameter_number": 1, "timestamps": ["2025-08-01 00:00:00+00:00",], "values": [</pre>
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			15.06597965738758] }
Send instructions to a device?	Don't have this currently		
Send additional data to sensor interface / FrugalIoT	Maybe some additional data processing? Analytics? To calculate something to do something?	POST We sent org ID, farm name and crop details	The partner sent an irrigation plan based on weather, crop and some other data.

We have a couple more APIs in the implementation but they are less relevant. We didn't store any sensor data in our db except the irrigation plan.

FarmOS

(Note - we at present don't have any direct contact with FarmOS - the following comes from Dan TerAvest at OurSci)

FarmOS has "Sensor" assets (to represent IoT devices and their location) and "Data Streams" to represent data of specific types. So for example, if you have a Raspberry Pi with a temperature and humidity sensor you can set up a Sensor asset for the Pi and 2 data streams (one for temp and one for humidity).

Process:

1. Users need to add the "Sensor Asset Module" to their FarmOS instance
2. From the "Assets" menu, select "sensor" to view all sensors. Select "Add Sensor Asset" to register a new sensor.
3. Add name, select if sensor needs private key or can be read publicly.
4. Add data stream: The data stream database storage is a plugin based system. The default plugin saves data to the farmOS database itself. But it's also possible to create a plugin that saves it somewhere else (like an InfluxDB or remote API).
5. Location:
 - a. If fixed location: then user enters the geometry of location
 - b. If not fixed, then geometry determined by movement logs
6. Can also add "ID Tag" information or other files (ex: image of asset).
7. Sensor data can then be accessed via the FarmOS API

Mitra: Does that data stream follow a FarmOS standard, or are there different plugins for different kinds of devices - in which case implementing a plugin to whatever standard we come up with

would be the way to go.

Greg: let's ask Mike! Post your questions to the farmOS forum, you'll get answers there quick

Other projects - not in contact yet

Field Kit

- (from Greg) [Fieldkit.org](https://fieldkit.org) - been around a while and widely used. I talked to them a long time ago and they have some very nice structures. I believe they are all open source also. Shah Selbe is the lead → shah@fieldkit.org.
- [Online community](#) - last activity aug 2024

Mitra: Looks interesting I'll reach out.

Mitra: I emailed but did not hear back, looking at the website the project may be dead

Home Assistant

- (from Greg) Many people are also using Home Assistant. Might be worth at least reviewing.
- Mitra says: I looked at Home Assistant when I started Frugal IoT, It made a whole bunch of assumptions that were reasonable for its context, but not elsewhere. They very much built around a home environment, with a local controller, rather than something like a farm environment with no local controller, and potentially not using WiFi (using GSM, or LoRa) etc . One simple example (if I remember correctly) was that OTA was pushed directly from the laptop to a local device on the same WiFi, rather than polled from a back-end.

AgStack

- agstack.org, Funded by Linux Foundation <https://github.com/agstack>
- Defunct, repos are still available but nothing remotely IoT as far as I (Mitra) could see.

ASABE - Ag Gateway

- <https://aggateway.org> Appears to be be “big-ag” and backend data interchange level rather than anything IoT that I could find.
-